



# HealthConnect Clinic

## Appointment Attendance and No-Show Analysis *Week 4 Initial Data Analytics Report*



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# CHAPTER 1

## INTRODUCTION

### 1.1 Business Context

HealthConnect Clinic is a fictional healthcare provider that manages appointment-based services. The clinic is experiencing several operational and patient-support challenges, including missed appointments, difficulty understanding patterns associated with no-shows, inefficient use of appointment capacity, repetitive patient enquiries, and the need to strengthen patient engagement and administrative support.

The central project question is:

**How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?**

Within this broader project, the Data Analytics track focuses on establishing the evidence base required to understand appointment attendance and no-show behavior. The Week 4 work therefore concentrates on understanding the available appointment data, assessing its quality, identifying relevant analytical variables, defining business questions, proposing meaningful KPIs, and establishing an initial analytical approach.

Week 4 is intended to provide a structured analytical foundation for subsequent stages of the HealthConnect project. It does not seek to deliver the complete analytical, predictive, or AI-supported solution.

### 1.2 Track-Specific Problem and Objective

The Data Analytics track addresses the need to establish a reliable evidence base for investigating **appointment attendance and no-show behavior** at HealthConnect Clinic.

The specific Week 4 objective is to determine whether the available appointment data provides a suitable foundation for subsequent analysis of attendance outcomes and to define how that analysis should be approached.

This involves:

Understanding the structure and content of the available dataset;

Assessing its completeness, consistency, and suitability for analysis;

Identifying variables that may be relevant to appointment attendance and no-shows;

Defining focused business questions;

Identifying and justifying potential KPIs; and

Establishing an initial analytical approach for subsequent project stages.

The objective at this stage is therefore **analytical planning and foundation-building**, rather than producing final findings or recommendations.

### 1.3 Role of the Data Analytics Track

The Data Analytics track provides the quantitative evidence foundation for the broader HealthConnect project. Its role is to transform the available appointment data into a structured analytical framework that can support subsequent investigation and decision-making.

Within the Week 4 stage, this role involves assessing the available data, identifying relevant variables and business questions, defining potential performance measures, and establishing an appropriate approach for analyzing appointment attendance and no-show patterns.

The outputs of this work are intended to support later stages of the project, including deeper analysis and, where justified by the evidence and project requirements, the development of predictive or AI-supported solutions.

The Data Analytics track therefore contributes by ensuring that subsequent interventions are informed by a clearly defined problem, appropriate measures, and an understanding of the strengths and limitations of the available data.

### 1.4 Scope of the Week 4 Analysis

The Week 4 analysis is limited to establishing the analytical foundation for investigating appointment attendance and no-show patterns at HealthConnect Clinic.

The scope includes:

Reviewing the appointment dataset and its structure;

Assessing data quality and identifying relevant data limitations;

Identifying variables that may be relevant to appointment outcomes;

Defining business questions for subsequent analysis;

Identifying and justifying 3–5 potential KPIs;

Establishing an initial analytical approach; and

Documenting assumptions, limitations, risks, dependencies, and other considerations that may affect the proposed approach.

The proposed analytical focus includes appointment outcomes and variables relating to patient history, reminder activity, booking behavior, scheduling characteristics, accessibility, and selected demographic characteristics.

At this stage, these variables are being identified because they may be relevant to the proposed business questions. Their relationships with appointment outcomes will be investigated only in subsequent analytical stages.



# CHAPTER 2

## DATASET OVERVIEW AND DATA QUALITY ASSESSMENT

### 2.1 Introduction

This chapter provides an overview of the HealthConnect appointment dataset and presents the initial assessment of its quality and suitability for the proposed Week 4 analysis. The purpose is to establish what information is available, how the dataset is structured, and whether any data-quality issues may affect subsequent analytical work.

The assessment focuses on the dataset's structure, identifiers, completeness, consistency, and basic logical relationships between related fields. It does not attempt to analyse appointment outcomes or establish relationships between variables. Those activities are reserved for the subsequent analytical stages defined in Chapters 3 and 5.

The findings in this chapter therefore provide the data foundation for identifying the relevant business questions and potential KPIs in the following chapters.

### 2.2 Dataset Overview

The HealthConnect dataset is provided as an Excel workbook containing 5,000 appointment records and 18 variables. The dataset is structured at the appointment-record level, with each row representing an individual scheduled appointment.

Each appointment is identified by an `appointment_id`, which serves as the primary appointment-level identifier. The dataset also contains an anonymised `patient_id`, meaning that the same patient may appear across multiple appointment records.

According to the accompanying data dictionary, the available information covers several broad categories:

- patient demographics;
- appointment characteristics;
- booking information;
- appointment scheduling;

- previous appointment history;
- reminder activity;
- accessibility;
- waiting time; and
- final appointment outcome.

The dataset is therefore sufficiently broad to support an initial investigation of appointment attendance and no-show behavior while also providing contextual variables that may be examined in subsequent stages.

*Table 2.1— HealthConnect Dataset Structure*

| <b>Dataset Characteristic</b> | <b>Description</b>                                           |
|-------------------------------|--------------------------------------------------------------|
| Source format                 | Excel workbook                                               |
| Number of records             | 5,000 appointment records                                    |
| Number of variables           | 18                                                           |
| Unit of analysis              | Individual appointment record                                |
| Appointment identifier        | appointment_id                                               |
| Patient identifier            | patient_id                                                   |
| Primary outcome field         | appointment_outcome                                          |
| Outcome categories            | Attended, No-Show, Cancelled                                 |
| Dataset type                  | Fictional and anonymised                                     |
| Primary analytical purpose    | Investigation of appointment attendance and no-show patterns |

The dataset's appointment-level structure is appropriate for the Week 4 objective because the central problem concerns whether scheduled appointments are attended or missed.

## 2.3 Dataset Variables

The dataset contains variables representing different aspects of the appointment process. At this stage, the variables are documented to establish the available analytical information; their specific relevance to individual business questions is addressed separately in Chapter 3.

*Table 2.2 — HealthConnect Dataset Fields and Definitions*

| Variable              | Data Type | Description                                    |
|-----------------------|-----------|------------------------------------------------|
| appointment_id        | Text      | Unique appointment identifier                  |
| patient_id            | Text      | Anonymised patient identifier                  |
| gender                | Text      | Recorded gender category                       |
| age                   | Integer   | Patient age in years                           |
| age_group             | Text      | Age band derived from age                      |
| appointment_type      | Text      | Type of scheduled appointment                  |
| booking_date          | Date      | Date on which the appointment was booked       |
| appointment_date      | Date      | Scheduled appointment date                     |
| appointment_day       | Text      | Day of the week of the appointment             |
| appointment_time      | Text      | Appointment time period                        |
| booking_lead_days     | Integer   | Number of days between booking and appointment |
| previous_appointments | Integer   | Number of previous appointments                |
| previous_no_shows     | Integer   | Number of previous no-shows                    |

| Variable              | Data Type | Description                                    |
|-----------------------|-----------|------------------------------------------------|
| reminder_sent         | Text      | Indicates whether a reminder was sent          |
| reminder_channel      | Text      | Channel through which a reminder was delivered |
| distance_to_clinic_km | Decimal   | Approximate distance from the clinic           |
| waiting_time_minutes  | Integer   | Estimated waiting time                         |
| appointment_outcome   | Text      | Final appointment status                       |

The data dictionary indicates that appointment\_day is derived from appointment\_date, while booking\_lead\_days is provided and can also be derived from the booking and appointment dates. These relationships should be retained and checked during subsequent data preparation rather than treating derived fields as independent sources of information.

The data dictionary also identifies the dataset as synthetic and anonymised, which is relevant when considering the scope and interpretation of the analysis.

## 2.4 Initial Data Quality Assessment

An initial data-quality assessment was conducted to determine whether the dataset provides a sufficiently reliable foundation for the proposed Week 4 analysis.

*Table 2.3—Initial Data Quality Assessment*

| Quality check                | Initial finding                                                                                                                                                           |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Appointment ID uniqueness    | 0 duplicate appointment_id values; 5,000 unique IDs across 5,000 records.                                                                                                 |
| Fully duplicated rows        | 0 fully duplicated rows identified.                                                                                                                                       |
| Reminder channel missingness | 1,366 missing values. The Dictionary states None is expected when no reminder was sent, so these values require contextual interpretation rather than automatic deletion. |
| Distance missingness         | 90 missing values; treatment should be documented before distance-based analysis.                                                                                         |
| Waiting-time missingness     | 60 missing values; treatment should be documented before analysis.                                                                                                        |
| Booking/appointment dates    | booking_date <= appointment_date for all records: True.                                                                                                                   |

|                       |                                                                                                        |
|-----------------------|--------------------------------------------------------------------------------------------------------|
| Booking lead time     | Provided booking_lead_days matches the date difference for all records: True.                          |
| Previous no-show rule | previous_no_shows <= previous_appointments for all records: True.                                      |
| Age                   | Observed age range: 18–80; age_group is derived from age and should be treated as a descriptive field. |
| Outcome values        | Observed categories: No-Show, Attended, Cancelled.                                                     |

The assessment considered:

- completeness;
- identifier uniqueness;
- duplicate records;
- data types;
- date consistency;
- logical consistency between related variables; and
- the nature of missing values.

The assessment indicates that the dataset is generally suitable for subsequent analysis, although several variables require specific documentation and treatment.

## 2.5 Completeness Assessment

Most of the dataset is complete. The core identification, demographic, appointment, booking, historical, reminder-status, and outcome fields contain no missing records.

The following variables were identified as complete across the 5,000 records:

- appointment\_id;
- patient\_id;
- gender;
- age;

- age\_group;
- appointment\_type;
- booking\_date;
- appointment\_date;
- appointment\_day;
- appointment\_time;
- booking\_lead\_days;
- previous\_appointments;
- previous\_no\_shows;
- reminder\_sent; and
- appointment\_outcome.

Three variables contain missing values.

*Table 2.4 — Missing-Value Assessment*

| Variable              | Missing Records | Initial Assessment                                                    |
|-----------------------|-----------------|-----------------------------------------------------------------------|
| reminder_channel      | 1,366           | Conditional/non-applicable missingness                                |
| distance_to_clinic_km | 90              | Limited missingness requiring documentation and appropriate treatment |
| waiting_time_minutes  | 60              | Limited missingness requiring documentation and appropriate treatment |

The missing values therefore do not have the same analytical meaning and should not be treated as a single generic missing-data problem.

### *2.5.1 Reminder\_channel*

The `reminder_channel` field contains 1,366 missing records. Importantly, these records correspond exactly to the 1,366 appointments for which `reminder_sent` is recorded as No.

This indicates that the blank reminder-channel values are likely structurally non-applicable rather than ordinary missing observations. Where no reminder was sent, there would be no reminder channel to record.

Consequently, these blank values should not automatically be replaced with an arbitrary category or treated as evidence that reminder-channel information was simply omitted.

For subsequent analysis, `reminder_channel` should therefore be interpreted together with `reminder_sent`.

### *2.5.2 Distance\_to\_clinic\_km*

The `distance_to_clinic_km` variable contains 90 missing records. This represents a relatively limited proportion of the dataset, but the missing observations should still be documented before the variable is used in subsequent analysis.

No assumption should be made about the missing distances without establishing an appropriate treatment strategy. Depending on the analytical method eventually selected, the records may need to be excluded from a specific analysis or handled using another justified approach.

### *2.5.3 Waiting\_time\_minutes*

The `waiting_time_minutes` variable contains 60 missing records. These observations should similarly be documented and treated consistently during subsequent analysis.

In addition to its missing values, this variable requires a separate temporal interpretation consideration. The data dictionary describes it as an estimated waiting time, but the available documentation does not establish sufficiently whether the measure is recorded before or after the appointment outcome is determined.

Therefore, it should not automatically be treated as a pre-attendance explanatory or predictive variable. Its appropriate analytical role will depend on confirmation of how and when the measure is recorded.

## 2.6 Duplicate and Identifier Assessment

The dataset contains 5,000 appointment records and 5,000 unique appointment\_id values.

No duplicate appointment identifiers were identified during the initial assessment.

This provides evidence that appointment\_id can be used as the unique appointment-level identifier for the proposed analysis. It also reduces the immediate risk of counting the same appointment more than once due to duplicated appointment identifiers.

The patient\_id field serves a different purpose. Because the data dictionary indicates that a patient identifier may appear across multiple appointments, it should not be expected to be unique at the appointment-record level.

## 2.7 Date Consistency

The relationship between booking\_date and appointment\_date was examined as part of the initial logical-consistency assessment.

No records were identified in which the booking date occurred after the scheduled appointment date.

This provides evidence that the date relationship is logically consistent for the initial analysis. It also supports the interpretation of booking\_lead\_days as the interval between booking and the scheduled appointment.

The relationship should nevertheless be retained as a validation rule during subsequent data preparation.

## 2.8 Historical Appointment Consistency

The relationship between previous\_appointments and previous\_no\_shows was also assessed.

No records were identified in which the number of previous no-shows exceeded the number of previous appointments.

This provides evidence that the two historical measures are internally consistent within the available data.



For later analysis, these variables should continue to be interpreted as historical information relating to appointments preceding the current appointment, consistent with their definitions in the dataset documentation.

## 2.9 Data-Type and Structural Considerations

The data dictionary identifies appropriate basic data types for the principal fields, including text identifiers and categorical fields, integers for count and age variables, decimal values for distance, and date fields for booking and appointment dates.

Several fields are also derived or closely related to other variables. For example:

- age\_group is derived from age;
- appointment\_day is derived from appointment\_date; and
- booking\_lead\_days can be derived from the difference between booking\_date and appointment\_date.

These relationships should be checked during subsequent data preparation to ensure that derived fields remain consistent with their source fields.

The presence of both supplied and derivable fields is not inherently a data-quality problem, but it creates a validation requirement. Any discrepancy between a derived field and its underlying source fields should be investigated before analysis.

## 2.10 Overall Data Quality Assessment

Based on the initial assessment, the HealthConnect dataset is considered generally suitable for the proposed Week 4 analytical work.

The principal strengths identified are:

- 5,000 appointment records;
- unique appointment identifiers;
- complete core appointment and outcome information;
- complete core demographic fields;
- logically consistent booking and appointment dates;

- logically consistent historical appointment measures; and
- clearly defined appointment outcome categories.

The principal data-quality considerations are:

- conditional missingness in reminder\_channel;
- limited missingness in distance\_to\_clinic\_km;
- limited missingness in waiting\_time\_minutes;
- the need to validate derived fields against their source fields; and
- the need to confirm the temporal meaning of waiting\_time\_minutes before assigning it a predictive or explanatory role.

These issues do not prevent the proposed Week 4 analysis. However, they should be explicitly documented and appropriately addressed before the variables are used in subsequent analytical stages.

## 2.11 Implications for Subsequent Analysis

The data-quality assessment establishes several conditions that should guide the next stages of the project.

First, appointment outcomes should be handled according to their defined categories. Attended, No-Show, and Cancelled should remain distinct outcomes unless a later analytical objective provides a documented reason for combining categories.

Second, reminder\_channel should be interpreted in conjunction with reminder\_sent because its missing values appear to represent non-applicability rather than random missingness.

Third, analyses involving distance\_to\_clinic\_km and waiting\_time\_minutes should account for their limited missing observations.

Fourth, any use of waiting\_time\_minutes in relation to appointment attendance should first establish when the measure is recorded. This is particularly important because a variable measured after attendance cannot appropriately be used as though it were information available before the attendance outcome.

Finally, the dataset should be preserved in its original form, with any subsequent cleaning or transformation decisions documented to maintain analytical traceability.

## 2.12 Summary

This chapter reviewed the structure and quality of the HealthConnect appointment dataset.

The dataset contains 5,000 appointment records and 18 variables covering appointment, patient, booking, historical, reminder, accessibility, waiting-time, and outcome information.

The initial data-quality assessment found the dataset to be generally suitable for the proposed analysis. The core identification and outcome fields are complete, appointment identifiers are unique, booking and appointment dates are logically consistent, and the historical appointment measures satisfy their basic consistency check.

The principal issues requiring attention are the conditional missingness in `reminder_channel`, limited missingness in `distance_to_clinic_km` and `waiting_time_minutes`, and the need to validate derived variables. In addition, the temporal definition of `waiting_time_minutes` must be established before the variable is assigned any predictive or pre-attendance explanatory role.

With these considerations documented, the dataset provides an appropriate foundation for the business questions and analytical priorities established in Chapter 3.

# CHAPTER 3

## ANALYTICAL VARIABLES AND BUSINESS QUESTIONS

### 3.1 Analytical Variable Framework

Following the data-quality assessment presented in Chapter 2, the dataset variables were classified according to their relevance to the investigation of appointment attendance and no-show patterns.

The variables are grouped into five analytical categories:

1. appointment outcome;
2. patient characteristics;
3. appointment and booking characteristics;
4. patient history and engagement; and
5. accessibility and operational experience.

This classification establishes which fields will be used as the primary outcome, which will be used for segmentation, and which may be examined as potential explanatory variables.

### 3.2 Appointment Outcome Variable

#### 3.2.1 *appointment\_outcome*

`appointment_outcome` is the primary outcome variable for the Data Analytics track because it records the result of each appointment.

The outcome categories are:

- Attended
- No-Show
- Cancelled

For the purposes of this analysis, these categories will remain analytically distinct.

In particular, **Cancelled** appointments will not automatically be classified as **No-Show** appointments. A cancellation represents a different patient behavior from failing to attend a scheduled appointment.

The primary analytical focus will therefore be on identifying patterns associated with **No-Show** outcomes while retaining Attended and Cancelled appointments as separate outcome categories.

### 3.3 Patient Characteristics

The following variables provide demographic characteristics that can be used to examine differences in appointment outcomes across patient segments.

#### 3.3.1 Gender

gender can be used to compare appointment outcomes across gender categories.

It should primarily be treated as a **segmentation variable** rather than being interpreted as a causal explanation for appointment attendance.

#### 3.3.2 Age

Age provides the patient's numerical age and can support analysis of attendance patterns across different age ranges.

#### 3.3.3 Age Group

Age\_group provides a categorical representation of age and can be used to compare no-show patterns between predefined age segments.

The use of both age and age\_group provides flexibility between numerical analysis and categorical segmentation.

### 3.4 Appointment and Booking Characteristics

The following variables describe when and how appointments were scheduled and therefore provide important contextual information for investigating attendance behavior.

#### *3.4.1 Appointment Type*

Appointment\_type identifies the type of appointment or service involved.

This variable can be used to determine whether no-show patterns differ across appointment types.

#### *3.4.2 Booking Date*

Booking\_date records when the appointment was booked.

It provides the starting point for examining booking behavior and calculating or validating the period between booking and the scheduled appointment.

#### *3.4.3 Appointment Date*

Appointment\_date identifies the scheduled date of the appointment.

It can be used to examine appointment timing and to derive temporal characteristics relevant to attendance analysis.

#### *3.4.4 Appointment Day*

appointment\_day identifies the day of the week on which the appointment is scheduled.

This variable can be used to investigate whether no-show patterns differ across days of the week.

#### *3.4.5 Appointment Time*

Appointment\_time identifies the scheduled time of the appointment.

It can be used to investigate whether appointment attendance differs across different periods of the day.

#### *3.4.6 Booking Lead Days*

Booking\_lead\_days measures the number of days between the booking date and the scheduled appointment date.

This is an important analytical variable because it can be used to investigate whether the length of time between booking and appointment is associated with attendance outcomes.

## 3.5 Patient History and Engagement

These variables provide information about previous patient behavior and the clinic's reminder activity.

### *3.5.1 Previous Appointments*

Previous\_appointments represents the patient's previous appointment history.

This variable can help distinguish patients with limited appointment history from those with a longer history of interaction with the clinic.

### *3.5.2 Previous No-Shows*

Previous\_no\_shows records the patient's historical no-show behavior.

This is particularly important because it allows the analysis to investigate whether previous no-show behavior is associated with subsequent appointment outcomes.

### *3.5.3 Reminder Sent*

Reminder\_sent indicates whether a reminder was sent for the appointment.

This variable is directly relevant to the investigation of the clinic's reminder process and its relationship with appointment attendance.

### *3.5.4 Reminder Channel*

Reminder\_channel identifies the communication channel used when a reminder was sent.

Because the data-quality assessment in Chapter 2 established that blank reminder-channel values correspond to appointments where no reminder was sent, this variable should be interpreted together with reminder\_sent.

The analysis should therefore avoid treating these conditional blanks as ordinary missing observations.

## 3.6 Accessibility and Operational Experience

Two variables provide additional information about accessibility and the patient experience.

### *3.6.1 Distance to Clinic*

Distance\_to\_clinic\_km provides an approximate measure of the distance between the patient and the clinic.

It may be used to investigate whether accessibility is associated with differences in appointment attendance.

However, the missing values identified in Chapter 2 must be considered when this variable is analyzed.

### *3.6.2 Waiting Time*

Waiting\_time\_minutes records the patient's waiting time.

This variable is primarily relevant to **operational performance and patient experience**.

It should therefore be distinguished from variables that are known before an appointment takes place. In particular, waiting time should not automatically be interpreted as a direct predictor of whether a patient will attend an appointment if the measure is recorded after arrival.

Its primary analytical value is likely to emerge when examining the broader patient-support and operational experience dimensions of the HealthConnect project.

## **3.7 Analytical Relevance of the Variables**

The variables can therefore be summarized according to their principal analytical role:



Table 3.1 Variables and their Analytical Roles

| Analytical category                      | Variables                                                                                              | Primary purpose                                          |
|------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Outcome                                  | appointment_outcome                                                                                    | Measure appointment results and identify no-shows        |
| Patient characteristics                  | gender, age, age_group                                                                                 | Demographic segmentation                                 |
| Appointment and booking                  | appointment_type, booking_date, appointment_date, appointment_day, appointment_time, booking_lead_days | Examine scheduling and booking patterns                  |
| Patient history and engagement           | previous_appointments, previous_no_shows, reminder_sent, reminder_channel                              | Examine historical behavior and reminder activity        |
| Accessibility and operational experience | distance_to_clinic_km, waiting_time_minutes                                                            | Examine accessibility and operational/patient experience |

This classification provides the analytical foundation for the business questions defined in the following section.

### 3.8 Business Questions

Based on the business problem, the dataset structure and the analytical variables identified above, five business questions have been established for the Week 4 analysis.

**BQ1 — What factors are associated with appointment no-shows at HealthConnect Clinic?**

This is the overarching analytical question.

It will guide the examination of appointment outcomes against relevant patient, booking, historical, reminder, scheduling and accessibility variables.

**BQ2 — Are patients with a history of previous no-shows more likely to miss subsequent appointments?**

This question focuses on whether historical attendance behavior is associated with future appointment outcomes.

Relevant variables include:

- previous\_appointments;
- previous\_no\_shows; and
- appointment\_outcome.

**BQ3 — Is appointment attendance associated with whether a patient receives a reminder?**

This question examines the relationship between reminder activity and appointment outcomes.

Relevant variables include:

- reminder\_sent;
- reminder\_channel; and
- appointment\_outcome.

The analysis will distinguish between appointments where a reminder was sent and those where no reminder was sent.

**BQ4 — Is booking lead time associated with appointment attendance and no-shows?**

This question investigates whether the amount of time between booking and the scheduled appointment is associated with different appointment outcomes.

The principal variable is:

- booking\_lead\_days.

The analysis will compare appointment outcomes across different booking-lead-time patterns.

### **BQ5 — Do no-show patterns differ by appointment type, day of the week, or time of day?**

This question focuses on scheduling-related patterns that may help HealthConnect understand whether certain appointment configurations are associated with higher or lower observed no-show rates.

Relevant variables include:

- appointment\_type;
- appointment\_day;
- appointment\_time; and
- appointment\_outcome.

## **3.9 Alignment Between Variables and Business Questions**

The relationship between the analytical variables and the business questions is summarized below.

*Table 3.2 —Business Questions and Connected Variables*

| <b>Business Question</b>                                     | <b>Primary variables</b>                               |
|--------------------------------------------------------------|--------------------------------------------------------|
| BQ1 — What factors are associated with appointment no-shows? | appointment_outcome and relevant explanatory variables |

| Business Question                                                     | Primary variables                                                        |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------|
| BQ2 — Does previous no-show behavior relate to subsequent attendance? | previous_appointments, previous_no_shows, appointment_outcome            |
| BQ3 — Is attendance associated with reminder activity?                | reminder_sent, reminder_channel, appointment_outcome                     |
| BQ4 — Is booking lead time associated with attendance?                | booking_lead_days, appointment_outcome                                   |
| BQ5 — Do scheduling characteristics relate to no-shows?               | appointment_type, appointment_day, appointment_time, appointment_outcome |

This alignment ensures that the subsequent KPI selection and analytical approach are based on clearly defined business questions rather than on exploratory analysis without a defined purpose.

### 3.10 Chapter Summary

This chapter classified the HealthConnect dataset variables according to their analytical roles and established five business questions for the Week 4 Data Analytics track.

Appointment\_outcome was identified as the principal outcome variable, with the remaining variables providing demographic, booking, scheduling, historical, reminder, accessibility and operational context.

The business questions focus primarily on factors associated with appointment no-shows, including previous no-show behavior, reminder activity, booking lead time and scheduling characteristics.

These questions provide the basis for the **proposed KPIs and initial analytical approach presented in Chapter 4.**

# CHAPTER 4

## POTENTIAL KPI FRAMEWORK

### 4.1 Chapter Introduction

This chapter identifies the potential Key Performance Indicators (KPIs) that may be used in subsequent stages of the HealthConnect Experience Lab project. The KPIs are derived from the business questions established in Chapter 3 and are intended to provide measurable indicators of appointment demand, attendance behavior, and service experience.

At the Week 4 stage, these KPIs are proposed measures only. They are not calculated, analyzed, or presented through visualizations. Their purpose is to establish a clear measurement framework that can guide the subsequent analytical work.

The selection of the KPIs is therefore based on three considerations: their relevance to the identified business questions, their ability to provide meaningful measures of operational performance, and the availability of appropriate variables within the dataset.

### 4.2 Potential KPI Framework

Four potential KPIs have been identified for further consideration. Each KPI is directly linked to a business question and has a defined analytical purpose.

*Table 4.1— Proposed KPI Framework*

| Potential KPI      | Linked business question                                | Why it is relevant                                                                                                                          |
|--------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 1. No-Show Rate    | What is the overall level of appointment no-shows?      | Establishes the baseline size of HealthConnect's missed-appointment problem and provides the main measure for tracking no-show performance. |
| 2. Attendance Rate | What proportion of scheduled appointments are attended? | Provides the complementary measure of successful attendance and helps assess how effectively scheduled                                      |

|                                        |                                                                                  |                                                                                                                                                                                            |
|----------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        |                                                                                  | appointment capacity is being used.                                                                                                                                                        |
| 3. Reminder-Associated No-Show Rate    | Is appointment attendance associated with whether a patient receives a reminder? | Directly evaluates whether no-show patterns differ between appointments where reminders were and were not sent.<br>Reminder channel can subsequently be examined as a secondary breakdown. |
| 4. Repeat No-Show Rate                 | Are patients with previous no-shows more likely to miss subsequent appointments? | Measures the relationship between historical no-show behaviour and future attendance, helping determine whether repeat no-show behaviour is an important pattern.                          |
| 5. No-Show Rate by Appointment Segment | Do no-show patterns differ by appointment type, day, time or booking lead time?  | Provides a consistent measure for comparing scheduling and booking segments and identifying appointment conditions associated with higher no-show levels.                                  |

## 4.3 Justification of the Proposed KPIs

### 4.3.1 No-Show Rate

No-Show Rate is proposed as the primary KPI because missed appointments represent the central behavioral outcome examined by the project. Expressing no-shows as a proportion of scheduled appointments provides a standardized measure that can subsequently be compared across relevant patient, appointment, and temporal segments.

The KPI can therefore support later analysis of whether no-show behavior varies across the dimensions identified in Chapter 3.

At this stage, however, the KPI is only being identified and justified. No numerical value or conclusion about the level of no-show performance is presented.

#### *4.3.2 Attendance Rate*

Attendance Rate provides a complementary perspective on appointment utilization. While No-Show Rate focuses specifically on missed appointments, Attendance Rate measures the proportion of scheduled appointments that result in attendance.

Using both measures in subsequent stages can provide a more complete representation of appointment outcomes. Their final interpretation will depend on the structure and completeness of the appointment outcome categories in the dataset.

#### *4.3.3 Reminder-Associated No-Show Rate*

*This KPI evaluates whether no-show patterns differ between appointments where reminders were sent and those where no reminder was sent. It directly addresses the business question concerning the relationship between reminder activity and appointment attendance. The KPI can also support further analysis by reminder channel, where appropriate.*

#### *4.3.4 Repeat No-Show Rate*

*This KPI measures the extent to which patients with a history of previous no-shows subsequently miss appointments. It directly addresses whether previous no-show behaviour is associated with future attendance and provides a focused measure of repeat non-attendance.*

#### *4.3.5 No-Show Rate by Appointment Segment*

*This KPI examines whether no-show patterns differ across appointment type, day of the week, time of day, or booking lead-time groups. It directly supports the analysis of scheduling and booking characteristics and can help identify segments with relatively higher or lower observed no-show rates for further investigation.*

If waiting time is recorded after the patient has attended, it should be treated as a service-experience measure rather than as a pre-attendance predictor. Accordingly, its use in analyzing attendance behavior will depend on confirmation of when the measure is recorded. No causal or predictive interpretation will be assigned to this variable at the Week 4 stage.

### **4.4 KPI-to-Business Question Alignment**

The proposed KPI framework is designed to maintain a direct connection between the business problem and subsequent analysis.

Table 4.2 — KPI and Business Question Alignment

| Business Question                                                                | Relevant KPI                        | Intended Use in Subsequent Stages                                                                                                                                       |
|----------------------------------------------------------------------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What is the overall level of appointment no-shows?                               | No-Show Rate                        | Establish the baseline size of the missed-appointment problem and compare no-show performance across relevant segments.                                                 |
| What proportion of scheduled appointments are attended?                          | Attendance Rate                     | Assess overall attendance performance and identify differences between relevant patient, appointment and scheduling groups.                                             |
| Is appointment attendance associated with whether a patient receives a reminder? | Reminder-Associated No-Show Rate    | Compare no-show patterns between appointments where reminders were and were not sent, with reminder channel examined as a secondary breakdown.                          |
| Are patients with previous no-shows more likely to miss subsequent appointments? | Repeat No-Show Rate                 | Examine the relationship between previous no-show behaviour and subsequent appointment attendance, and identify whether repeat no-shows represent an important pattern. |
| Do no-show patterns differ by appointment type, day, time or booking lead time?  | No-Show Rate by Appointment Segment | Compare no-show performance across appointment and scheduling segments and identify conditions associated with higher no-show levels.                                   |

This alignment ensures that each proposed KPI has a clear analytical purpose rather than being included simply because the corresponding variable is available in the dataset.



## 4.5 Scope of KPI Development at Week 4

The KPI framework represents the measurement design stage of the project. The following activities are intentionally deferred to subsequent stages:

- calculation of KPI values;
- comparison of KPI performance across groups;
- identification of high- or low-performing segments;
- visualization of KPI results;
- statistical testing of KPI relationships; and
- development of predictive or prescriptive models based on the KPIs.

This approach is consistent with the Week 4 requirement that KPIs should be identified and their relevance justified, while calculation, analysis, and visualization are reserved for subsequent stages of the project.

## 4.6 Chapter Summary

This chapter has identified four potential KPIs relevant to the HealthConnect business questions: No-Show Rate, Attendance Rate, Appointment Volume, and Average Waiting Time.

Each KPI has been linked to a specific business question and justified according to its potential analytical and operational value. The framework deliberately avoids calculating or interpreting KPI results at this stage, in accordance with the Week 4 scope.

The proposed measures provide a focused foundation for the initial analytical approach presented in Chapter 5, where the methods for investigating the identified business questions will be established.

# CHAPTER 5

## INITIAL ANALYSIS APPROACH

### 5.1 Chapter Introduction

This chapter outlines the proposed approach for analyzing the HealthConnect appointment dataset in subsequent stages of the project. The approach is based on the business questions identified in Chapter 3 and the potential KPIs established in Chapter 4.

At the Week 4 stage, the purpose is to define how the data will be analysed, rather than to present completed analytical results. KPI calculations, detailed statistical analysis, visualisations, and predictive modelling are therefore outside the scope of this chapter.

The proposed approach combines descriptive analysis, comparative analysis, segmentation, and, where appropriate, temporal analysis. The methods will be applied according to the structure and quality of the available data.

### 5.2 Descriptive Analysis

The first stage of the proposed analysis will be to establish a clear understanding of the appointment population and its recorded outcomes.

Descriptive analysis will examine:

The overall volume of appointments;

The distribution of appointment outcomes;

The characteristics of relevant categorical variables;

The distribution of relevant numerical variables; and

The extent of missing or incomplete information affecting the variables selected for further analysis.

For categorical variables, frequencies and percentages will be considered. For numerical variables, appropriate measures such as the mean, median, minimum, maximum, and measures of dispersion may be used where relevant.

This stage will provide the baseline required for subsequent comparisons and will help determine whether particular variables are suitable for further analysis.

### 5.3 Comparative and Segmentation Analysis

The second stage will examine whether appointment outcomes differ across relevant groups identified through the business questions and dataset assessment.

Potential comparison dimensions may include relevant demographic characteristics, appointment characteristics, scheduling-related variables, and other variables identified as analytically appropriate.

The principal outcome of interest will be appointment attendance behavior, particularly the distinction between attended appointments and no-shows.

Where appropriate, the proposed analysis will compare:

Appointment volumes;

Attendance rates;

No-show rates; and

Relevant distributions or summary statistics.

The purpose of this analysis will be to identify differences and associations between groups, rather than to establish causal relationships. Segment sizes will also be considered so that conclusions are not based solely on percentages from very small groups.

### 5.4 Temporal Analysis

Where the dataset contains sufficiently reliable date or time information, temporal analysis will be considered to determine whether appointment patterns vary across available time periods.

Potential dimensions may include:

Appointment date;

Day of the week;

Month;

Scheduling period; and

Other meaningful time intervals supported by the available data.

The analysis may examine changes in appointment volume and attendance-related measures over time.

Temporal analysis will remain dependent on the completeness, consistency, and appropriate definition of the available date fields. Any observed variation across time will be interpreted as a descriptive pattern rather than evidence of a causal effect.

## 5.5 Application of the Proposed KPIs

The four potential KPIs identified in Chapter 4 will provide the measurement framework for subsequent stages:

No-Show Rate — to assess the prevalence of missed appointments.

Attendance Rate — to assess the proportion of appointments resulting in attendance.

Appointment Volume — to establish the scale of appointment activity.

Average Waiting Time — to assess the observed waiting-time experience, subject to validation of the variable definition.

During Week 4, these KPIs are only being identified and justified. Their numerical calculation and analytical interpretation will be undertaken in subsequent stages if supported by the dataset and project requirements.

## 5.6 Treatment of waiting\_time\_minutes

The variable `waiting_time_minutes` requires specific methodological consideration before it is incorporated into further analysis.

Its analytical role depends on when the waiting time is measured relative to the appointment outcome. If the variable is recorded after a patient arrives or after attendance has occurred, it cannot appropriately be treated as a pre-attendance predictor of whether that patient will attend.

Accordingly, the proposed approach is to initially treat `waiting_time_minutes` as a potential service-experience variable. Its use in any analysis of attendance behavior will depend on confirmation from the dataset documentation that its measurement occurs before the relevant outcome.

This precaution helps avoid temporal leakage and prevents a post-outcome variable from being incorrectly interpreted as an antecedent factor.

## 5.7 Analytical Decision Criteria

The selection of analytical techniques in subsequent stages will be guided by the characteristics of the data rather than by the availability of a particular analytical tool.

Before applying a method, consideration will be given to:

The measurement type of the variable;

Data completeness;

Distribution and variability;

Sample size within comparison groups;

Consistency of category definitions;

The temporal ordering of variables where relevant; and

The specific business question being addressed.

More advanced statistical or predictive methods will only be considered where they provide a clear analytical benefit and are appropriate for the available data.

## 5.8 Proposed Analytical Sequence

The proposed analysis will follow a progressive sequence:

Data validation → Descriptive analysis → KPI calculation → Comparative/segmentation analysis → Temporal analysis where appropriate → Interpretation

This sequence ensures that subsequent analysis is grounded in an understanding of the dataset and that more detailed comparisons are not undertaken without first establishing the underlying data structure.

The sequence may be adjusted if further investigation reveals data-quality, technical, or methodological constraints.

## 5.9 Chapter Summary

This chapter has established the initial analytical approach for the HealthConnect Experience Lab project. The proposed approach will begin with descriptive analysis before progressing to comparative, segmentation, and, where appropriate, temporal analysis.

The potential KPIs identified in Chapter 4 will provide the measurement framework for subsequent stages, while their calculation and interpretation remain outside the scope of the Week 4 submission.

Particular attention will be given to the definition and temporal position of `waiting_time_minutes` to ensure that it is not incorrectly treated as a pre-attendance predictor. Overall, the proposed approach provides a structured pathway for progressing from the Week 4 analytical assessment toward deeper analysis in subsequent stages.

# CHAPTER 6

## ASSUMPTIONS, LIMITATIONS, RISKS AND DEPENDENCIES

### 6.1 Chapter Introduction

The proposed analytical approach is subject to several assumptions, limitations, risks, and dependencies that may affect the quality, interpretation, and implementation of the analysis. Identifying these factors at the initial stage helps ensure that subsequent analytical work remains realistic and appropriately cautious.

The considerations presented in this chapter complement the data-quality assessment in Chapter 2 and the methodological considerations established in Chapters 3–5. They are intended to identify factors that may influence the proposed approach rather than to provide conclusions about the final analytical results.

### 6.2 Assumptions

The proposed analysis is based on the following assumptions:

1. **Appointment records are sufficiently representative of the population being examined.** The available dataset is assumed to provide an appropriate basis for the initial investigation of appointment attendance and no-show behavior.
2. **Appointment outcome categories are meaningfully defined.** It is assumed that the recorded attendance and no-show categories represent the intended operational outcomes and can be distinguished consistently.
3. **Relevant variables are recorded consistently.** Variables selected for subsequent analysis are assumed to have sufficiently consistent definitions and formats to support meaningful comparison.
4. **Date-related fields, where available, represent the intended events.** Any temporal analysis will depend on confirmation that the relevant date fields correspond to the events they are assumed to represent.
5. **The dataset contains sufficient information for the identified business questions.** The initial approach assumes that the available variables provide a reasonable starting

point for investigating appointment attendance, while recognizing that additional information may be required for deeper analysis.

### 6.3 Data Limitations

The analysis may be affected by limitations inherent in the available dataset.

First, the dataset may not capture all factors that influence appointment attendance. Variables relating to patient circumstances, communication effectiveness, accessibility, operational disruptions, or other contextual factors may be unavailable.

Second, the analysis is based on observational data. Consequently, observed relationships between variables cannot automatically be interpreted as causal relationships.

Third, the usefulness of subgroup comparisons may depend on the number of observations available within each group. Small groups may produce unstable rates and should therefore be interpreted cautiously.

Finally, the analytical value of individual variables depends on the accuracy and consistency of their definitions. Variables whose meaning is unclear will require validation before being used in advanced analysis.

### 6.4 Missing or Incomplete Information

Missing or incomplete information may affect both the reliability and coverage of the proposed analysis.

Missing values may reduce the number of observations available for particular analyses and may also affect comparisons between groups if missingness is not distributed evenly.

Before subsequent analysis, important variables should therefore be assessed for:

- missing values;
- inconsistent categories;
- invalid values;
- duplicate or potentially duplicated records; and
- unexpected data formats.



Where missing information materially affects an analysis, the limitation should be documented rather than concealed through unsupported assumptions or inappropriate imputation.

## 6.5 Technical Constraints

The proposed analysis may be influenced by the tools and technical environment available during subsequent stages.

Potential technical considerations include:

- compatibility between the source dataset and the selected analytical tools;
- limitations in data-processing or statistical software;
- data-format and transformation requirements;
- computational resources available for more advanced analysis; and
- the ability to reproduce analytical calculations consistently.

These constraints are not expected to prevent the initial analysis but may influence the complexity or implementation of later analytical techniques.

## 6.6 Project Risks

Several risks could affect the quality or interpretation of the project.

### *6.6.1 Misinterpretation of Variables*

A variable may be incorrectly interpreted if its definition or measurement process is unclear. This is particularly relevant when determining whether a variable can legitimately be used to explain or predict an appointment outcome.

### *6.6.2 Temporal Leakage*

Using information recorded after an appointment outcome to predict or explain that outcome could introduce temporal leakage. This risk is particularly relevant to `waiting_time_minutes` and will need to be addressed before any predictive interpretation is attempted.

### *6.6.3 Overinterpretation of Associations*

Observed differences between groups may be incorrectly interpreted as causal relationships. The project will therefore distinguish between descriptive patterns, associations, and causal explanations.

### *6.6.4 Small-Sample Comparisons*

Very small subgroups may produce apparently high or low rates that are not sufficiently stable for strong conclusions. Subgroup size will therefore be considered alongside percentage-based measures.

### *6.6.5 Scope Creep*

There is a risk of introducing advanced statistical, predictive, or visualization activities before the foundational analytical questions have been adequately addressed. The project will therefore follow the staged approach defined in this report.

## **6.7 System and Project Dependencies**

The proposed analytical approach depends on several factors being available or confirmed during subsequent stages.

These include:

- access to the complete and relevant dataset;
- reliable definitions of key fields;
- appropriate analytical software and computing resources;
- confirmation of the meaning and timing of important variables;
- sufficient data quality for the intended analytical techniques; and
- availability of any additional contextual information required to interpret important findings.

If any of these dependencies change, the analytical approach may need to be revised.

## 6.8 Modelling Considerations

At the Week 4 stage, no predictive model is being proposed as a completed analytical output. However, any future modelling activity should consider the temporal and structural characteristics of the dataset.

In particular, variables used to predict appointment attendance should be available **before the outcome occurs**. Variables recorded after attendance or non-attendance should not be treated as valid predictive inputs.

Future modelling should also consider issues such as:

- outcome definition;
- class imbalance, if present;
- missing data;
- feature availability at prediction time;
- model interpretability;
- validation methodology; and
- the operational purpose of any resulting model.

These considerations will become relevant only if predictive analysis forms part of a later project stage.

## 6.9 Ethical and Safety Considerations

Appointment datasets may contain information relating to individuals and therefore require responsible handling.

Subsequent analysis should avoid unnecessary disclosure of personally identifiable information and should use aggregated results wherever possible for reporting and presentation.

The analysis should also avoid using demographic or other sensitive characteristics to make unsupported judgments about individuals or groups. Where group differences are identified, they should be treated as analytical observations requiring appropriate contextual interpretation rather than as evidence of individual behavior or responsibility.

Any future predictive modelling should additionally consider the potential for unfair or discriminatory outcomes, particularly if variables associated with demographic or socioeconomic characteristics are included.

## 6.10 Chapter Summary

This chapter has identified the principal assumptions, limitations, risks, and dependencies that may affect the proposed HealthConnect analysis.

The main considerations include data completeness, variable definitions, subgroup size, technical constraints, temporal leakage, potential overinterpretation of associations, and the availability of reliable information at the appropriate point in time.

Particular caution is required when interpreting `waiting_time_minutes`, as its suitability for predictive analysis depends on when the measure is recorded relative to the appointment outcome.

Recognizing these considerations at Week 4 provides a stronger foundation for subsequent analysis and helps ensure that future findings are technically appropriate, transparent, and proportionate to the evidence available.

# CHAPTER 7

## WEEK 4 PROJECT SUMMARY

### 7.1 The Problem Addressed

The HealthConnect Experience Lab project focuses on understanding appointment attendance and no-show behavior within the available appointment dataset. The Week 4 work establishes the analytical foundation required to investigate the scale of missed appointments, identify relevant patterns, and determine which measures can support subsequent decision-making.

The focus at this stage is not to provide final findings or interventions, but to define a clear and evidence-based direction for the subsequent analysis.

### 7.2 Resources Used

The Week 4 analysis was developed using the available HealthConnect appointment dataset and the supporting project requirements and documentation.

The principal resources used were:

- the HealthConnect appointment dataset;
- the available dataset field definitions and metadata;
- analytical and data-quality assessment techniques;
- the Week 4 project requirements; and
- the analytical framework developed during the project.

These resources were used to assess the dataset, identify relevant business questions, establish potential KPIs, and develop the proposed analytical approach.

### 7.3 Key Observations from Week 4

The Week 4 assessment established several important observations.

First, the dataset provides a basis for examining appointment attendance and no-show behavior through the recorded appointment outcomes.

Second, the analysis requires a clear distinction between appointment volume, attendance performance, and no-show performance. This supports the selection of focused KPIs rather than relying on isolated counts or individual variables.

Third, the proposed analysis should focus on differences and associations across relevant patient, appointment, and temporal dimensions while avoiding unsupported causal interpretations.

Finally, the interpretation of `waiting_time_minutes` requires particular caution. Its analytical role depends on when the variable is recorded relative to the appointment outcome. It should therefore not automatically be treated as a pre-attendance predictor.

## 7.4 Proposed Approach

The proposed analytical approach is to begin with descriptive analysis and data validation before progressing to comparative and segmentation analysis. Where reliable date information is available, temporal analysis may also be undertaken.

Four potential KPIs have been identified for subsequent stages:

1. No-Show Rate
2. Attendance Rate
3. Appointment Volume
4. Average Waiting Time

At Week 4, these KPIs have only been identified and justified. Their calculation, analysis, and visualization will be considered in subsequent project stages.

## 7.5 Key Considerations

The main factors that may affect subsequent work include data completeness, variable definitions, missing or inconsistent information, subgroup sizes, technical constraints, and the temporal ordering of variables.

The project must also avoid treating observed associations as causal relationships. Any future predictive analysis must ensure that input variables are available before the outcome being predicted.

Responsible handling of appointment data will also be required, particularly where records may contain information relating to identifiable individuals.

## 7.6 Proposed Focus for Week 5

The proposed focus for Week 5 is to move from the analytical framework established in Week 4 toward implementation of the initial analysis.

Subject to the project requirements and dataset readiness, Week 5 should focus on:

- validating the variables selected for analysis;
- implementing the proposed data preparation steps;
- calculating the approved KPIs;
- conducting the initial descriptive and comparative analysis;
- examining relevant attendance and no-show patterns; and
- documenting the resulting findings clearly and objectively.

The methodological caution concerning `waiting_time_minutes` should be retained during this stage so that the variable is used according to its documented timing and analytical role.

## 7.7 Chapter Summary

Week 4 established the analytical foundation for investigating HealthConnect appointment attendance and no-show behavior. The work identified the relevant business problem, assessed the available data, established four potential KPIs, defined an initial analytical approach, and documented the assumptions, limitations, risks, and dependencies that may affect subsequent work.

The proposed Week 5 focus is to move from analytical planning to initial implementation and analysis, while maintaining the methodological and data-quality safeguards established during Week 4.